

Assignment-3.2

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1)

Code:

```
def calculator():
```

```
    print("Simple Calculator")
```

```
    while True:
```

```
        # Input numbers with error handling
```

```
        try:
```

```
            num1 = float(input("Enter first number: "))
```

```
            num2 = float(input("Enter second number: "))
```

```
        except ValueError:
```

```
            print("Error: Please enter a valid number.")
```

```
            continue
```

```
        # Input operator
```

```
        operator = input("Enter operator (+, -, *, /): ").strip()
```

```
        # Perform calculation
```

```
        if operator == '+':
```

```
            result = num1 + num2
```

```
        elif operator == '-':
```

```
            result = num1 - num2
```

```
elif operator == '*':  
    result = num1 * num2  
elif operator == '/':  
    if num2 != 0:  
        result = num1 / num2  
    else:  
        print("Error: Division by zero is not allowed.")  
        continue  
else:  
    print("Error: Invalid operator.")  
    continue  
print(f"Result: {result}")  
# Ask if the user wants to continue  
again = input("Do you want to perform another calculation?  
(y/n): ").strip().lower()  
if again != 'y':  
    print("Goodbye!")  
    break  
# Run the calculator  
calculator()
```

Output:

```
/ai_coding/New folder/ass_3.py"
Simple Calculator
Enter first number: 2
Enter second number: 3
Enter operator (+, -, *, /): +
Result: 5.0
Do you want to perform another calculation? (y/n): n
Goodbye!
```

2)

Code:

```
def sort_students(students):
    # Sort by marks (descending), then by name (ascending)
    return sorted(students, key=lambda x: (-x[1], x[0]))

students = []

n = int(input("Enter number of students: "))

for i in range(n):
    name = input(f"Enter name of student {i+1}: ")
    marks = int(input(f"Enter marks of student {i+1}: "))
    students.append((name, marks))

sorted_students = sort_students(students)

print("\nSorted Student List:")

for name, marks in sorted_students:
    print(name, marks)
```

Output:

```
Enter number of students: 4
Enter name of student 1: ananya
Enter marks of student 1: 98
Enter name of student 2: pooja
Enter marks of student 2: 98
Enter name of student 3: hasini
Enter marks of student 3: 40
Enter name of student 4: nandini
Enter marks of student 4: 20

Sorted Student List:
ananya 98
pooja 98
hasini 40
nandini 20
```

3)

Code:

```
def is_prime(n):
    # Handle edge cases
    if n <= 1:
        return False
    if n == 2:
        return True
    if n % 2 == 0:
        return False
    # Check divisibility up to sqrt(n)
    for i in range(3, int(n ** 0.5) + 1, 2):
        if n % i == 0:
```

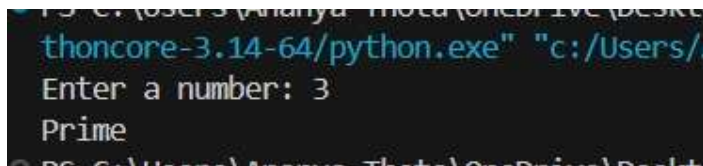
```
        return False

    return True

# Taking input from user
num = int(input("Enter a number: "))

if is_prime(num):
    print("Prime")
else:
    print("Not Prime")
```

Output:



```
thoncore-3.14-64/python.exe "c:/Users/...
Enter a number: 3
Prime
```

4)

Code:

```
def calculate_grade(percentage):
    if percentage >= 90:
        return "A+"
    elif percentage >= 80:
        return "A"
    elif percentage >= 70:
        return "B"
```

```
elif percentage >= 60:
    return "C"
elif percentage >= 50:
    return "D"
else:
    return "Fail"

print("==== Student Grading System ====")
name = input("Enter Student Name: ")
roll_no = input("Enter Roll Number: ")
subjects = int(input("Enter number of subjects: "))
total_marks = 0
max_marks = subjects * 100
for i in range(subjects):
    marks = int(input(f"Enter marks for subject {i+1}: "))
    total_marks += marks
percentage = (total_marks / max_marks) * 100
grade = calculate_grade(percentage)
print("\n==== Result ====")
print("Name    :", name)
print("Roll No  :", roll_no)
print("Total Marks:", total_marks, "/", max_marks)
print("Percentage :", round(percentage, 2), "%")
print("Grade    :", grade)
```

Output:

```
===== Student Grading System =====
Enter Student Name: janaki
Enter Roll Number: 234
Enter number of subjects: 4
Enter marks for subject 1: 90
Enter marks for subject 2: 80
Enter marks for subject 3: 70
Enter marks for subject 4: 60

===== Result =====
Name      : janaki
Roll No   : 234
Total Marks: 300 / 400
Percentage : 75.0 %
Grade     : B
```

5)

Code:

```
def km_to_miles(km):
    return km * 0.621371

def miles_to_km(miles):
    return miles / 0.621371

print("=== Unit Conversion System ===")
print("1. Kilometers to Miles")
print("2. Miles to Kilometers")
choice = int(input("Enter your choice (1 or 2): "))
if choice == 1:
    km = float(input("Enter distance in kilometers: "))
    print("Distance in miles:", round(km_to_miles(km), 2))
```

```
elif choice == 2:
```

```
    miles = float(input("Enter distance in miles: "))
```

```
    print("Distance in kilometers:", round(miles_to_km(miles), 2))
```

```
else:
```

```
    print("Invalid choice")
```

Output:

```
=== Unit Conversion System ===
1. Kilometers to Miles
2. Miles to Kilometers
Enter your choice (1 or 2): 1
Enter distance in kilometers: 40
Distance in miles: 24.85
```